1.Create a comparison table listing the main differences between DDR3, DDR4, and DDR5 RAM, including speed, voltage, and typical use cases.

Feature	DDR3 RAM	DDR4 RAM	DDR5 RAM
Full form	Double Data Rate 3	Double Data Rate 4	Double Data Rate 5
Speed	800–2133 MT/s (typical range)	1600–3200 MT/s (standard range)	4800–6400+ MT/s
Voltage	1.5 V	1.2 V	1.1 V
Performance	Low	Medium to high	High
Power consumption	Higher	Lower than DDR3	Lower voltage, higher bandwidth
Typical use	Older computers	Most older and mid-range modern PCs	Newer high-performance PCs and workstations

**2.Open your laptop or desktop (with power disconnected), locate the RAM slots, and take a clear photo of the installed RAM module. Reinstall the RAM module and verify that your system boots up successfully.

Hint: Refer to your device manual or a trusted YouTube guide if unsure about RAM removal and installation steps.**

1. Shut down the laptop or desktop completely.
2. Disconnect the power cable and other connected devices. For a laptop, disconnect the internal battery if the manufacturer's instructions require it and it is safe to do so.
3. Open the computer's RAM compartment or back cover using the device manual.
4. Locate the RAM slot and take a clear photo of the installed RAM module.
5. Release the retaining clips or screws as appropriate and carefully remove the module.
6. Reinstall the RAM in the correct orientation and make sure it is securely seated.
7. Close the cover, reconnect power, and start the computer.
8. Check that the system boots and the installed memory is detected in BIOS/UEFI or the operating system.

3.List three key differences between HDD, SSD, and NVMe storage types, and for each, name one app or use case (from apps you use daily) that would benefit most from that storage type.

Feature	HDD	SATA SSD	NVMe SSD
Technology	Spinning magnetic disks	Flash memory	Flash memory using NVMe protocol
Speed	Slow	Faster than HDD	Usually the fastest
Moving parts	Yes	No	No
Connection	Usually SATA	Usually SATA	Usually PCIe through an M.2 slot
Cost per GB	Usually low	Medium	Usually higher
Daily app/use case	Storing movies, photos, and backups	Opening documents and using Microsoft Word	Booting Windows, opening VS Code, or running virtual machines

**4.Remove the storage drive (HDD or SSD) from your system, take a photo of the connector type (SATA, M.2, or NVMe), and reinstall the drive. Confirm that your device recognizes the storage after reassembly.

Constraint: Only perform this task if you have access to a device you are allowed to open. If not, find and share a YouTube video link showing the process for your laptop/PC model.**

1. Shut down the computer and disconnect the power supply.
2. Follow the manufacturer's service manual to open the laptop or desktop safely.
3. Locate the storage drive and take a clear photo showing its connector and label.
4. Carefully remove the retaining screw or connector, as applicable.
5. Reinstall the drive securely in its original position.
6. Reassemble the computer and reconnect the power.
7. Start the system and verify that the drive is detected in BIOS/UEFI or the operating system.